

Práctica 1: Instalación y uso de un clúster Hadoop con integración Hive y Sqoop

Objetivo

Esta práctica tiene como objetivo montar un entorno funcional de Big Data con Hadoop utilizando Docker, ejecutar tareas de procesamiento distribuidas, trabajar con herramientas de consulta como Hive y realizar integraciones de datos mediante Sqoop. Además, se comprobarán aspectos como la replicación y el crecimiento del sistema.

Parte 1: Instalación del clúster Hadoop

Crear un fichero docker-compose.yml que contenga:

- NameNode
- ResourceManager
- Dos DataNodes
- NodeManager
- Servicios de Prometheus y Grafana para la monitorización

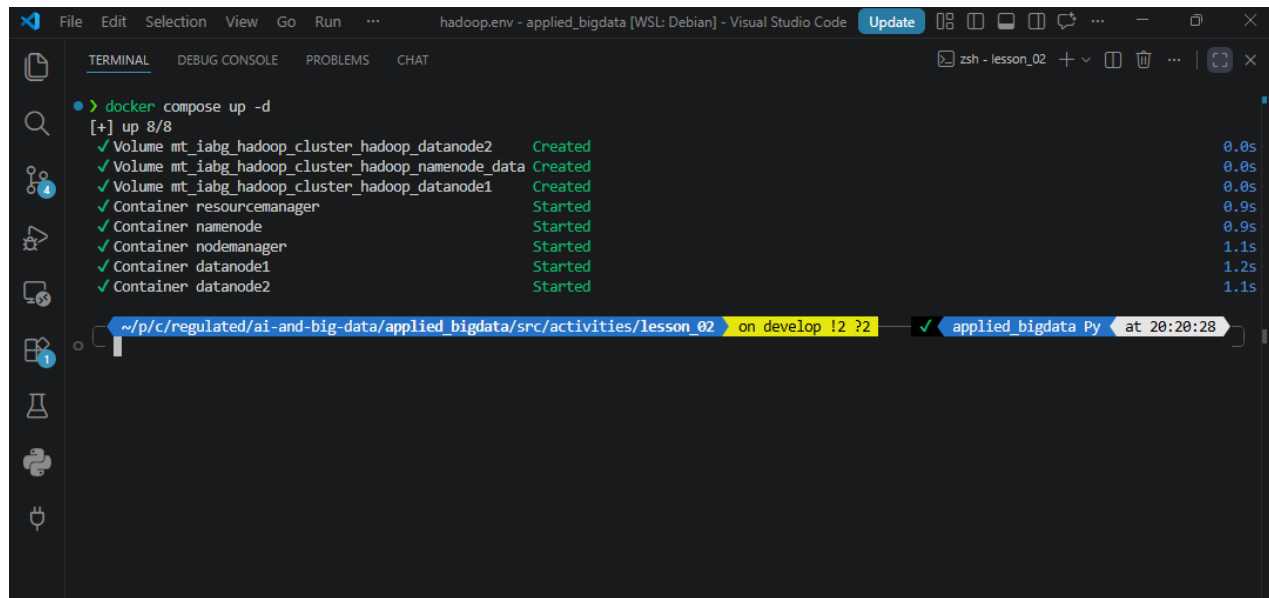
docker-compose.yml

```
name: mt_iabg_hadoop_cluster
services:
  namenode:
    image: bde2020/hadoop-namenode:2.0.0-hadoop3.2.1-java8
    container_name: namenode
    restart: always
    ports:
      - "9870:9870"
      - "9000:9000"
    env_file:
      - ./hadoop.env
    volumes:
      - hadoop_namenode_data:/hadoop/dfs/name

  resourcemanager:
    image: bde2020/hadoop-resourcemanager:2.0.0-hadoop3.2.1-java8
    container_name: resourcemanager
    hostname: resourcemanager
    restart: always
    ports:
      - "8088:8088"
    env_file:
      - ./hadoop.env

  datanode1:
    image: bde2020/hadoop-datanode:2.0.0-hadoop3.2.1-java8
    container_name: datanode1
    hostname: datanode1
    restart: always
    depends_on:
      - namenode
    env_file:
```

```
- ./hadoop.env
volumes:
- hadoop_datanode1:/hadoop/dfs/data
datanode2:
  image: bde2020/hadoop-datanode:2.0.0-hadoop3.2.1-java8
  container_name: datanode2
  hostname: datanode2
  restart: always
  depends_on:
  - namenode
  env_file:
  - ./hadoop.env
  volumes:
  - hadoop_datanode2:/hadoop/dfs/data
nodemanager:
  image: bde2020/hadoop-nodemanager:2.0.0-hadoop3.2.1-java8
  container_name: nodemanager
  hostname: nodemanager
  restart: always
  depends_on:
  - resourcemanager
  env_file:
  - ./hadoop.env
volumes:
  hadoop_namenode_data:
  hadoop_datanode1:
  hadoop_datanode2:
```



```
File Edit Selection View Go Run ... hadoop.env - applied_bigdata [WSL: Debian] - Visual Studio Code Update
TERMINAL DEBUG CONSOLE PROBLEMS CHAT zsh - lesson_02
> docker compose up -d
[+] up 8/8
✓ Volume mt_iabg_hadoop_cluster_hadoop_datanode2 Created 0.0s
✓ Volume mt_iabg_hadoop_cluster_hadoop_namenode_data Created 0.0s
✓ Volume mt_iabg_hadoop_cluster_hadoop_datanode1 Created 0.0s
✓ Container resourcemanager Started 0.9s
✓ Container namenode Started 0.9s
✓ Container nodemanager Started 1.1s
✓ Container datanode1 Started 1.2s
✓ Container datanode2 Started 1.1s
~/p/c/regulated/ai-and-big-data/applied_bigdata/src/activities/lesson_02 on develop !2.?? ✓ applied_bigdata.py at 20:20:28
```

Docker Desktop > Containers

The screenshot shows the Docker Desktop interface with the 'Containers' tab selected. The left sidebar lists various Docker features. The main area displays a table of running containers for a Hadoop cluster. The table includes columns for Name, Container ID, Image, Port(s), CPU (%), Last started, and Actions. Below the table, there are 'Walkthroughs' for 'Multi-container applications' and 'Containerize your application'.

Name	Container ID	Image	Port(s)	CPU (%)	Last started	Actions
mt_labg_hadoop_cluster	-	-	-	2.02%	3 minutes ago	[Stop] [Refresh] [Delete]
resourcemanager	dadf3794a82a	bde2020/hadoop-resourcemanager:2.0.0-hado	8088:8088	0.88%	3 minutes ago	[Stop] [Refresh] [Delete]
namenode	c0d45eeb004d	bde2020/hadoop-namenode:2.0.0-hadoop3.2.1	9000:9000	0.34%	3 minutes ago	[Stop] [Refresh] [Delete]
datanode1	813bacd580b0	bde2020/hadoop-datanode:2.0.0-hadoop3.2.1	Show all ports (2)	0.23%	3 minutes ago	[Stop] [Refresh] [Delete]
nodemanager	2bedd7f3c723	bde2020/hadoop-nodemanager:2.0.0-hadoop3		0.4%	3 minutes ago	[Stop] [Refresh] [Delete]
datanode2	da718d071cec	bde2020/hadoop-datanode:2.0.0-hadoop3.2.1		0.17%	3 minutes ago	[Stop] [Refresh] [Delete]

Showing 6 items

Walkthroughs:

- Multi-container applications (8 mins)
- Containerize your application (3 mins)

View more in the Learning center

Engine running | RAM 6.63 GB | CPU 0.25% | Disk: 7.67 GB used (limit 1006.85 GB) | Terminal | Update available

Docker Desktop > Volumes

The screenshot shows the Docker Desktop interface with the 'Volumes' tab selected. The left sidebar lists various Docker features. The main area displays a table of volumes for the Hadoop cluster. The table includes columns for Name, Created, Size, and Actions. Below the table, there are 'Walkthroughs' for 'Multi-container applications' and 'Persist your data between containers'.

Name	Created	Size	Actions
mt_labg_hadoop_cluster_hadoop_datanode1	3 minutes ago	548 Bytes	[Refresh] [Delete]
mt_labg_hadoop_cluster_hadoop_datanode2	3 minutes ago	548 Bytes	[Refresh] [Delete]
mt_labg_hadoop_cluster_hadoop_namenode_data	3 minutes ago	1 MB	[Refresh] [Delete]

Showing 3 items

Walkthroughs:

- Multi-container applications (8 mins)
- Persist your data between containers (3 mins)

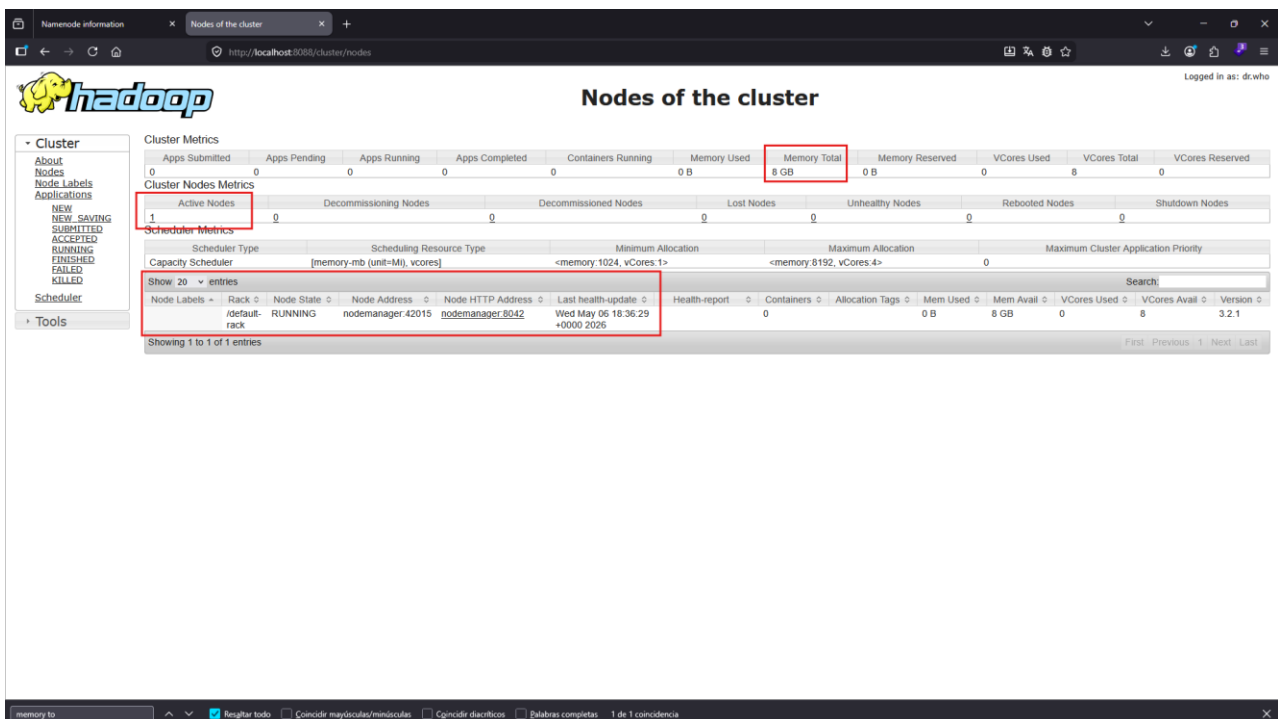
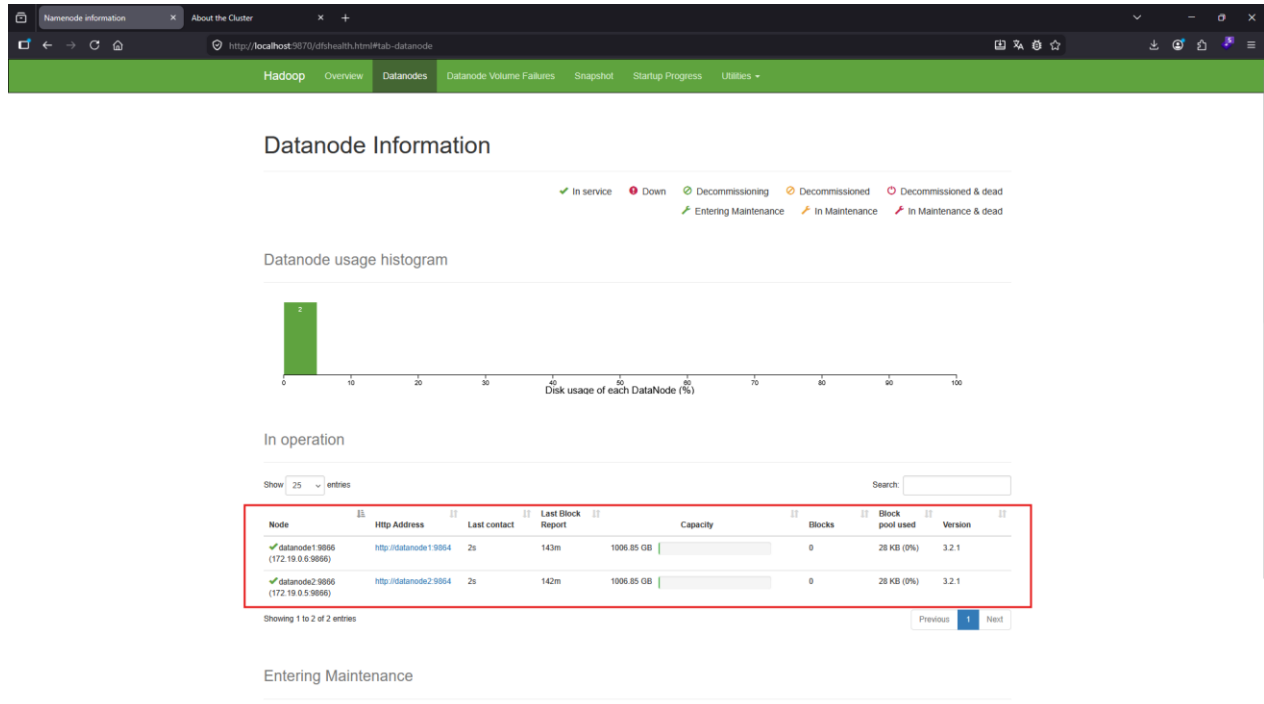
View more in the Learning center

Engine running | RAM 6.64 GB | CPU 0.03% | Disk: 7.67 GB used (limit 1006.85 GB) | Terminal | Update available

[David Cueli] Comentarios

Despliegue con **docker compose** levantando una infraestructura que incluye *NameNode*, *ResourceManager*, dos *DataNodes* (1 y 2) y el *NodeManager* para crear un entorno distribuido que simula un clúster real de Big Data en un solo host.

- Verificar desde la interfaz web (puertos 9870 para el NameNode y 8088 para YARN) que el sistema reconoce todos los nodos.



[David Cueli] Comentarios

Se accede a los paneles de control de Hadoop y YARN para confirmar, mediante interfaz web, que todos los nodos están activos y verificar que el clúster está listo para recibir trabajos

- Comprobar la replicación de archivos entre nodos.

```

modified:   src/activities/lesson_02/hadoop.env
new file:   src/temp.js
new file:   src/temp.json

• git commit -m "Getting the Hadoop cluster up with the docker-compose.yml and hadoop.env configuration files"
[develop 066bb8a] Getting the Hadoop cluster up with the docker-compose.yml and hadoop.env configuration files
4 files changed, 3115 insertions(+), 12 deletions(-)
create mode 100644 src/temp.js
create mode 100644 src/temp.json

• git push
Enumerating objects: 15, done.
Counting objects: 100% (15/15), done.
Delta compression using up to 88 threads
Compressing objects: 100% (9/9), done.
Writing objects: 100% (9/9), 8.86 KiB | 2.68 MiB/s, done.
Total 9 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To github.com:dcuelli-com/applied.bigdata.git
   b2cb011..066bb8a develop -> develop

• docker exec -it namenode /bin/bash
root@cd45eeb004d:/# echo "Test of replication" > replication.test.txt
root@cd45eeb004d:/# ls -la
total 424
drwxr-xr-x 1 root root 4096 May  8 16:05 .
drwxr-xr-x 1 root root 4096 May  8 16:05 ..
-rwxr-xr-x 1 root root  0 May  6 18:20 dockerenv
-rw-r--r-- 1 root root 325083 Feb  4 2020 KEYS
drwxr-xr-x 1 root root 4096 Feb  4 2020 bin
drwxr-xr-x 2 root root 4096 Sep  8 2019 boot
drwxr-xr-x 5 root root 340 May  8 15:25 dev
-rwxr-xr-x 1 root root 4155 Feb  4 2020 entrypoint.sh
drwxr-xr-x 1 root root 4096 May  6 18:20 etc
drwxr-xr-x 3 root root 4096 Feb  4 2020 hadoop
drwxr-xr-x 2 root root 4096 Feb  4 2020 hadoop-data
drwxr-xr-x 2 root root 4096 Sep  8 2019 home
drwxr-xr-x 1 root root 4096 Jan 30 2020 lib
drwxr-xr-x 2 root root 4096 Jan 30 2020 lib64
drwxr-xr-x 2 root root 4096 Jan 30 2020 media
drwxr-xr-x 2 root root 4096 Jan 30 2020 mnt
drwxr-xr-x 1 root root 4096 Feb  4 2020 opt
drwxr-xr-x 922 root root  0 May  8 15:25 proc
-rw-r--r-- 1 root root  22 May  8 16:05 replication.test.txt
drwx----- 1 root root 4096 Feb  4 2020 root
drwxr-xr-x 3 root root 4096 Jan 30 2020 run
-rwxr-xr-x 1 root root 494 Feb  4 2020 run.sh
drwxr-xr-x 1 root root 4096 Feb  4 2020/sbin
drwxr-xr-x 2 root root 4096 Jan 30 2020/srv
drwxr-xr-x 13 root root  0 May  8 15:25 sys
drwxrwxrwt 1 root root 4096 May  8 15:25 tmp
drwxr-xr-x 1 root root 4096 Jan 30 2020/usr
drwxr-xr-x 1 root root 4096 Jan 30 2020/var
root@cd45eeb004d:/# cat replication.test.txt
Test of replication
root@cd45eeb004d:/# hdfs dfs -put replication.test.txt /
2026-05-08 16:07:07,324 INFO sasl.SaslDataTransferClient: SASL encryption trust check: localhostTrusted = false, remoteHostTrusted = false
root@cd45eeb004d:/# hdfs fsck /replication.test.txt -files -blocks -locations
Connecting to namenode via http://namenode:9870/fsck?ugi=root&files=1&blocks=1&locations=1&path=/replication.test.txt
FSCK started by root (auth:SIMPLE) from /172.19.0.6 for path /replication.test.txt at Fri May 08 16:08:36 UTC 2026
/replication.test.txt: 20 bytes, replicated: replication-2, 1 block(s): OK
0. BP-268666459-172.19.0.2-1778091631795:blk_1073741825_1001 len=20 Live repl=2 [DataNodeInfoWithStorage[172.19.0.5:9866 DS-23f0912c-dad4-490
7-bae1-84d272c3d894, DiskInfoWithStorage[172.19.0.3:9866 DS-1a0a3596-7fc1-4578-bcef-1278429df12f, DISK]]]

Status: HEALTHY
Number of data-nodes: 2
Number of racks: 1
Total dirs: 0
Total symlinks: 0

Replicated Blocks:
Total size: 20 B
Total files: 1
Total blocks (validated): 1 (avg. block size 20 B)
Minimally replicated blocks: 1 (100.0 %)
Over-replicated blocks: 0 (0.0 %)
Under-replicated blocks: 0 (0.0 %)
Mis-replicated blocks: 0 (0.0 %)
Default replication factor: 2
Average block replication: 2.0
Missing blocks: 0
Corrupt blocks: 0
Missing replicas: 0 (0.0 %)

Erasure Coded Block Groups:
Total size: 0 B
Total files: 0
Total block groups (validated): 0
Minimally erasure-coded block groups: 0
Over-erasure-coded block groups: 0
Under-erasure-coded block groups: 0
Unsatisfactory placement block groups: 0
Average block group size: 0.0
Missing block groups: 0
Corrupt block groups: 0
Missing internal blocks: 0
FSCK ended at Fri May 08 16:08:36 UTC 2026 in 8 milliseconds

The filesystem under path '/replication.test.txt' is HEALTHY
root@cd45eeb004d:/#

```

[David Cueli] Comentarios

Se sube un archivo y se confirma ubicación para asegurar que Hadoop cumple con su función copiando los bloques de datos en diferentes DataNodes según el factor de replicación definido.

Parte 2: Ejecución de un trabajo MapReduce

- Crear un archivo de texto grande desde el terminal del NameNode (por ejemplo, con un bucle).
- Subirlo a HDFS.

```
root@c0d45eeb004d:/# for i in {1..1000}; do echo "hadoop mapreduce Bigdata example" >> dcc_big_file.txt; done
root@c0d45eeb004d:/# hdfs dfs -mkdir -p /user/root/input
root@c0d45eeb004d:/# hdfs dfs -put dcc_big_file.txt /user/root/input
2026-05-08 18:19:24,336 INFO sasl.SaslDataTransferClient: SASL encryption trust check: localhostTrusted = false,
remoteHostTrusted = false
root@c0d45eeb004d:/# hdfs dfs -ls /user/root/input
Found 1 items
-rw-r--r--  2 root supergroup      165000 2026-05-08 18:19 /user/root/input/dcc_big_file.txt
root@c0d45eeb004d:/#
```

[David Cueli] Comentarios

Genero un archivo de texto de “*gran tamaño*” mediante un bucle en el NameNode y lo cargo en HDFS para disponer de un volumen de datos para probar MapReduce.

- Ejecutar el trabajo wordcount para procesar el archivo.

```
root@c0d45eeb004d:/# hadoop jar /opt/hadoop-3.2.1/share/hadoop/mapreduce/hadoop-mapreduce-examples-3.2.1.jar wordcount
/user/root/input /user/root/output
```

```
2026-05-08 18:29:30,683 INFO Impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
2026-05-08 18:29:30,752 INFO Impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
2026-05-08 18:29:30,753 INFO Impl.MetricsSystemImpl: JobTracker metrics system started
2026-05-08 18:29:30,454 INFO InputFileOutputFormat: Total input files to process = 1
2026-05-08 18:29:30,478 INFO mapreduce.JobSubmitter: number of splits=1
2026-05-08 18:29:30,523 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_local351052453_0001
2026-05-08 18:29:30,523 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-05-08 18:29:30,433 INFO mapreduce.Job: The url to track the job: http://localhost:8080/
2026-05-08 18:29:30,434 INFO mapreduce.Job: Running Job: job_local351052453_0001
2026-05-08 18:29:30,436 INFO mapreduce.LocalJobRunner: OutputCommitter set in config null
2026-05-08 18:29:30,445 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-05-08 18:29:30,445 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup_temporary folders under output directory:false, ign
one cleanup failures: false
2026-05-08 18:29:30,446 INFO mapred.LocalJobRunner: OutputCommitter is org.apache.hadoop.mapreduce.lib.output.FileOutputCommitter
2026-05-08 18:29:30,500 INFO mapred.LocalJobRunner: Waiting for map tasks
2026-05-08 18:29:30,500 INFO mapred.LocalJobRunner: Starting task: attempt_local351052453_0001_r_000000_0
2026-05-08 18:29:30,522 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-05-08 18:29:30,522 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup_temporary folders under output directory:false, ign
one cleanup failures: false
2026-05-08 18:29:30,540 INFO mapred.Task: Using ResourceCalculatorProcessTree: []
2026-05-08 18:29:30,544 INFO mapred.Task: hdfs://name0:9000/user/root/input/dc_big_file.txt@1650800
2026-05-08 18:29:30,570 INFO mapred.MapTask: (EQATOR) 0 kv1 26214396(104857584)
2026-05-08 18:29:30,570 INFO mapred.MapTask: soft limit at 41888000
2026-05-08 18:29:30,570 INFO mapred.MapTask: bufstart = 0, bufused = 104857600
2026-05-08 18:29:30,570 INFO mapred.MapTask: kstart = 26214396, length = 655360
2026-05-08 18:29:30,580 INFO mapred.MapTask: Map output collector class = org.apache.hadoop.mapred.MapTask$MapOutputBuffer
2026-05-08 18:29:30,611 INFO org.apache.hadoop.mapred.Task: SAG encryption trust check: localHostTrusted = false, remoteHostTrusted = false
2026-05-08 18:29:30,731 INFO mapred.LocalJobRunner: map
2026-05-08 18:29:30,731 INFO mapred.LocalJobRunner: Starting flush of map output
2026-05-08 18:29:30,734 INFO mapred.MapTask: Spilling map output
2026-05-08 18:29:30,734 INFO mapred.MapTask: bufstart = 0, bufend = 245800; bufused = 104857600
2026-05-08 18:29:30,740 INFO mapred.MapTask: kstart = 26214396(104857584); kused = 26134408(104537500); length = 79907/655360
2026-05-08 18:29:30,760 INFO mapred.MapTask: Finished split 0
2026-05-08 18:29:30,799 INFO mapred.Task: Task:attempt_local351052453_0001_r_000000_0 is done. And is in the process of committing
2026-05-08 18:29:30,804 INFO mapred.Task: Task: attempt_local351052453_0001_r_000000_0 done.
2026-05-08 18:29:30,814 INFO mapred.Task: Final Counters for attempt_local351052453_0001_r_000000_0: Counters: 24
File System Counters
FILE: Number of bytes read=116779
FILE: Number of bytes written=439574
FILE: Number of read operations=0
FILE: Number of large read operations=0
FILE: Number of write operations=0
HDFS: Number of bytes read=165080
HDFS: Number of bytes written=0
HDFS: Number of read operations=5
HDFS: Number of large read operations=0
HDFS: Number of write operations=1
HDFS: Number of bytes read erasure-coded=0
Map-Reduce Framework
Map input records=5000
Map output records=20000
Map output bytes=245800
Map output materialized bytes=63
Input split bytes=118
Combine input records=20000
Combine output records=4
Spilled Records=4
Failed Shuffles=0
Merged Map outputs=0
GC time elapsed (ms)=0
Total committed heap usage (bytes)=1440219136
File Input Format Counters
Bytes Read=165080
2026-05-08 18:29:30,815 INFO mapred.LocalJobRunner: Finishing task: attempt_local351052453_0001_r_000000_0
2026-05-08 18:29:30,815 INFO mapred.LocalJobRunner: map task executor complete.
2026-05-08 18:29:30,819 INFO mapred.LocalJobRunner: Waiting for reduce tasks
2026-05-08 18:29:30,820 INFO mapred.LocalJobRunner: Starting task: attempt_local351052453_0001_r_000000_0
2026-05-08 18:29:30,820 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-05-08 18:29:30,820 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup_temporary folders under output directory:false, ign
one cleanup failures: false
2026-05-08 18:29:30,830 INFO mapred.Task: Using ResourceCalculatorProcessTree: []
2026-05-08 18:29:30,834 INFO mapred.ReduceTask: Using ShuffleConsumerPlugin: org.apache.hadoop.mapreduce.task.reduce.Shuffle@2a0f7527
2026-05-08 18:29:30,835 INFO Impl.MetricsSystemImpl: JobTracker metrics system already initialized
2026-05-08 18:29:30,857 INFO reduce.MergeManagerImpl: The max number of bytes for a single in-memory shuffle cannot be larger than Integer.MAX_
VALUE. Setting it to Integer.MAX_VALUE
2026-05-08 18:29:30,858 INFO reduce.MergeManagerImpl: MergeManager: memoryLimit=20041957376, maxSingleShuffleLimit=2147483647, mergeThreshold
=1121703812, latestSector=0, mergeWaitingOutputSizeThreshold=10
2026-05-08 18:29:30,861 INFO reduce.EventFetcher: attempt_local351052453_0001_r_000000_0 Thread started: EventFetcher for fetching Map Comple
tion Events
2026-05-08 18:29:30,884 INFO reduce.LocalFetcher: localFetcher#1 about to shuffle output of map attempt_local351052453_0001_r_000000_0 decomp
: 59 len: 63 to MEMORY
2026-05-08 18:29:30,888 INFO reduce.InMemoryMapOutput: Read 59 bytes from map-output for attempt_local351052453_0001_r_000000_0
2026-05-08 18:29:30,889 INFO reduce.MergeManagerImpl: CloseInMemoryFile -> map-output of size: 59, InMemoryMapOutputs.size() -> 1, commitMemor
y -> 0, usedMemory -> 59
2026-05-08 18:29:30,890 INFO reduce.EventFetcher: EventFetcher is interrupted.. Returning
2026-05-08 18:29:30,891 INFO mapred.LocalJobRunner: 1 / 1 copied.
2026-05-08 18:29:30,891 INFO reduce.MergeManagerImpl: FinalMerge called with 1 in-memory map-outputs and 0 on-disk map-outputs
2026-05-08 18:29:30,898 INFO mapred.Merger: Merging 1 sorted segments
2026-05-08 18:29:30,899 INFO mapred.Merger: Done to the last merge-pass, with 1 segments left of total size: 49 bytes
2026-05-08 18:29:30,900 INFO reduce.MergeManagerImpl: Merging 1 sorted segments
2026-05-08 18:29:30,900 INFO reduce.MergeManagerImpl: Merging 1 files, 63 bytes from disk
2026-05-08 18:29:30,900 INFO reduce.MergeManagerImpl: Merging 0 segments, 0 bytes from memory into reduce
2026-05-08 18:29:30,902 INFO mapred.Merger: Merging 1 sorted segments
2026-05-08 18:29:30,902 INFO reduce.Merger: Done to the last merge-pass, with 1 segments left of total size: 49 bytes
2026-05-08 18:29:30,902 INFO mapred.LocalJobRunner: 1 / 1 copied.
2026-05-08 18:29:30,937 INFO Configuration.deprecation: mapred.skip.on is deprecated. Instead, use mapreduce.job.skiprecords
2026-05-08 18:29:30,970 INFO org.apache.hadoop.mapred.Task: SAG encryption trust check: localHostTrusted = false, remoteHostTrusted = false
2026-05-08 18:29:31,025 INFO mapred.Task: Task:attempt_local351052453_0001_r_000000_0 is done. And is in the process of committing
2026-05-08 18:29:31,029 INFO mapred.LocalJobRunner: 1 / 1 copied.
2026-05-08 18:29:31,029 INFO mapred.Task: Task: attempt_local351052453_0001_r_000000_0 is allowed to commit now
2026-05-08 18:29:31,045 INFO output.FileOutputCommitter: Saved output of task 'attempt_local351052453_0001_r_000000_0' to hdfs://name0:9000
/user/root/output
2026-05-08 18:29:31,046 INFO mapred.LocalJobRunner: reduce > reduce
2026-05-08 18:29:31,046 INFO mapred.Task: Task: attempt_local351052453_0001_r_000000_0 done.
2026-05-08 18:29:31,048 INFO mapred.Task: Final Counters for attempt_local351052453_0001_r_000000_0: Counters: 30
File System Counters
FILE: Number of bytes read=116877
FILE: Number of bytes written=439597
FILE: Number of read operations=0
FILE: Number of large read operations=0
FILE: Number of write operations=0
HDFS: Number of bytes read=165080
HDFS: Number of bytes written=0
HDFS: Number of read operations=5
HDFS: Number of large read operations=0
HDFS: Number of write operations=3
HDFS: Number of bytes read erasure-coded=0
Map-Reduce Framework
Combine input records=0
Combine output records=0
Reduce input records=4
Reduce input groups=4
Reduce shuffle bytes=63
Reduce output records=4
Reduce output groups=4
Spilled Records=4
Shuffled Merges=1
Failed Shuffles=0
Merged Map outputs=1
GC time elapsed (ms)=0
Total committed heap usage (bytes)=1440219136
Shuffle Errors
BAD_ID=0
CONNECTION=0
IO_ERROR=0
MRIO_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0
File Output Format Counters
Bytes Written=53
2026-05-08 18:29:31,048 INFO mapred.LocalJobRunner: Finishing task: attempt_local351052453_0001_r_000000_0
2026-05-08 18:29:31,048 INFO mapred.LocalJobRunner: reduce task executor complete.
2026-05-08 18:29:31,446 INFO mapreduce.Job: Job: job_local351052453_0001_ran in 4m 40s: false
2026-05-08 18:29:31,442 INFO mapreduce.Job: map 100% reduce 100%
2026-05-08 18:29:31,443 INFO mapreduce.Job: Job: job_local351052453_0001 completed successfully
2026-05-08 18:29:31,450 INFO mapreduce.Job: mapreduce.job.counters=30
File System Counters
FILE: Number of bytes read=633596
FILE: Number of bytes written=1679131
FILE: Number of read operations=0
FILE: Number of large read operations=0
FILE: Number of write operations=0
HDFS: Number of bytes read=338000
HDFS: Number of bytes written=53
HDFS: Number of read operations=15
HDFS: Number of large read operations=0
HDFS: Number of write operations=4
HDFS: Number of bytes read erasure-coded=0
Map-Reduce Framework
Map input records=5000
Map output records=20000
Map output bytes=245800
Map output materialized bytes=63
Input split bytes=118
Combine input records=20000
Combine output records=4
Reduce input groups=4
Reduce shuffle bytes=63
Reduce input records=4
Reduce output records=4
Spilled Records=4
Shuffled Merges=1
Failed Shuffles=0
Merged Map outputs=1
GC time elapsed (ms)=0
Total committed heap usage (bytes)=2080438272
Shuffle Errors
BAD_ID=0
CONNECTION=0
IO_ERROR=0
MRIO_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0
File Output Format Counters
Bytes Written=53
File Output Format Counters
Bytes Written=53
```

```
2026-05-08 18:29:30,888 INFO reduce.InMemoryMapOutput: Read 59 bytes from map-output for attempt_local351052453_0001_r_000000_0
2026-05-08 18:29:30,889 INFO reduce.MergeManagerImpl: CloseInMemoryFile -> map-output of size: 59, InMemoryMapOutputs.size() -> 1, commitMemor
y -> 0, usedMemory -> 59
2026-05-08 18:29:30,890 INFO reduce.EventFetcher: EventFetcher is interrupted.. Returning
2026-05-08 18:29:30,891 INFO mapred.LocalJobRunner: 1 / 1 copied.
2026-05-08 18:29:30,891 INFO reduce.MergeManagerImpl: FinalMerge called with 1 in-memory map-outputs and 0 on-disk map-outputs
2026-05-08 18:29:30,898 INFO mapred.Merger: Merging 1 sorted segments
2026-05-08 18:29:30,899 INFO mapred.Merger: Done to the last merge-pass, with 1 segments left of total size: 49 bytes
2026-05-08 18:29:30,900 INFO reduce.MergeManagerImpl: Merging 1 sorted segments
2026-05-08 18:29:30,900 INFO reduce.MergeManagerImpl: Merging 1 files, 63 bytes from disk
2026-05-08 18:29:30,901 INFO reduce.Merger: Merging 0 segments, 0 bytes from memory into reduce
2026-05-08 18:29:30,902 INFO mapred.Merger: Merging 1 sorted segments
2026-05-08 18:29:30,902 INFO reduce.Merger: Done to the last merge-pass, with 1 segments left of total size: 49 bytes
2026-05-08 18:29:30,902 INFO mapred.LocalJobRunner: 1 / 1 copied.
2026-05-08 18:29:30,937 INFO Configuration.deprecation: mapred.skip.on is deprecated. Instead, use mapreduce.job.skiprecords
2026-05-08 18:29:30,970 INFO org.apache.hadoop.mapred.Task: SAG encryption trust check: localHostTrusted = false, remoteHostTrusted = false
2026-05-08 18:29:31,025 INFO mapred.Task: Task:attempt_local351052453_0001_r_000000_0 is done. And is in the process of committing
2026-05-08 18:29:31,029 INFO mapred.LocalJobRunner: 1 / 1 copied.
2026-05-08 18:29:31,029 INFO mapred.Task: Task: attempt_local351052453_0001_r_000000_0 is allowed to commit now
2026-05-08 18:29:31,045 INFO output.FileOutputCommitter: Saved output of task 'attempt_local351052453_0001_r_000000_0' to hdfs://name0:9000
/user/root/output
2026-05-08 18:29:31,046 INFO mapred.LocalJobRunner: reduce > reduce
2026-05-08 18:29:31,046 INFO mapred.Task: Task: attempt_local351052453_0001_r_000000_0 done.
2026-05-08 18:29:31,048 INFO mapred.Task: Final Counters for attempt_local351052453_0001_r_000000_0: Counters: 30
File System Counters
FILE: Number of bytes read=116877
FILE: Number of bytes written=439597
FILE: Number of read operations=0
FILE: Number of large read operations=0
FILE: Number of write operations=0
HDFS: Number of bytes read=165080
HDFS: Number of bytes written=0
HDFS: Number of read operations=5
HDFS: Number of large read operations=0
HDFS: Number of write operations=3
HDFS: Number of bytes read erasure-coded=0
Map-Reduce Framework
Combine input records=0
Combine output records=0
Reduce input records=4
Reduce input groups=4
Reduce shuffle bytes=63
Reduce output records=4
Reduce output groups=4
Spilled Records=4
Shuffled Merges=1
Failed Shuffles=0
Merged Map outputs=1
GC time elapsed (ms)=0
Total committed heap usage (bytes)=1440219136
Shuffle Errors
BAD_ID=0
CONNECTION=0
IO_ERROR=0
MRIO_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0
File Output Format Counters
Bytes Written=53
2026-05-08 18:29:31,048 INFO mapred.LocalJobRunner: Finishing task: attempt_local351052453_0001_r_000000_0
2026-05-08 18:29:31,048 INFO mapred.LocalJobRunner: reduce task executor complete.
2026-05-08 18:29:31,446 INFO mapreduce.Job: Job: job_local351052453_0001_ran in 4m 40s: false
2026-05-08 18:29:31,442 INFO mapreduce.Job: map 100% reduce 100%
2026-05-08 18:29:31,443 INFO mapreduce.Job: Job: job_local351052453_0001 completed successfully
2026-05-08 18:29:31,450 INFO mapreduce.Job: mapreduce.job.counters=30
File System Counters
FILE: Number of bytes read=633596
FILE: Number of bytes written=1679131
FILE: Number of read operations=0
FILE: Number of large read operations=0
FILE: Number of write operations=0
HDFS: Number of bytes read=338000
HDFS: Number of bytes written=53
HDFS: Number of read operations=15
HDFS: Number of large read operations=0
HDFS: Number of write operations=4
HDFS: Number of bytes read erasure-coded=0
Map-Reduce Framework
Map input records=5000
Map output records=20000
Map output bytes=245800
Map output materialized bytes=63
Input split bytes=118
Combine input records=20000
Combine output records=4
Reduce input groups=4
Reduce shuffle bytes=63
Reduce input records=4
Reduce output records=4
Spilled Records=4
Shuffled Merges=1
Failed Shuffles=0
Merged Map outputs=1
GC time elapsed (ms)=0
Total committed heap usage (bytes)=2080438272
Shuffle Errors
BAD_ID=0
CONNECTION=0
IO_ERROR=0
MRIO_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0
File Output Format Counters
Bytes Written=53
File Output Format Counters
Bytes Written=53
```

[David Cueli] Comentarios

Lanzo el JAR de ejemplos para contar palabras y alidar que MapReduce funciona correctamente.

- Visualizar los resultados del procesamiento en HDFS.

```

root@cd45eeb004d:/# hdfs dfs -ls /user/root/output
Found 2 items
-rw-r--r--  2 root supergroup      0 2026-05-08 18:29 /user/root/output/_SUCCESS
-rw-r--r--  2 root supergroup    53 2026-05-08 18:29 /user/root/output/part-r-00000
root@cd45eeb004d:/# hdfs dfs -cat /user/root/output/part-r-00000
2026-05-08 18:37:57.152 INFO sasl.SaslDataTransferClient: SASL encryption trust check: localhostTrusted = false, remoteHostTrusted = false
Bigdata 5000
example 5000
hadoop 5000
mapreduce 5000
root@cd45eeb004d:/#

```

[David Cueli] Comentarios

Consulto el contenido del directorio de salida en HDFS para confirmar que el proceso ha finalizado con éxito y que los resultados son los esperados.

Parte 3: Integración con Hive

```

# Parte 3: Integración con Hive
hive-server:
  image: bde2020/hive:2.3.2-postgresql-metastore
  container_name: hive-server
  depends_on:
    namenode:
      condition: service_started
    db-psql-metastore:
      condition: service_healthy
  env_file:
    - ./hadoop.env
  environment:
    - "HIVE_SITE_CONF_javax_jdo_option_ConnectionURL=jdbc:postgresql://db-psql
metastore:5432/metastore_db"
    - "HIVE_SITE_CONF_javax_jdo_option_ConnectionDriverName=org.postgresql.Driver"
    - "HIVE_SITE_CONF_javax_jdo_option_ConnectionUserName=hive"
    - "HIVE_SITE_CONF_javax_jdo_option_ConnectionPassword=mine"
  ports:
    - "10000:10000"
  volumes:
    - hive_metastore:/opt/hive/metastore_db
# Como hay RAM de sobra, indico un límite de memoria
deploy:
  resources:
    limits:
      memory: 8G

```

[David Cueli] Comentarios

Instalación de Hive en un contenedor de Docker con una base de datos de Postgres para los metadatos, guardando la estructura de las tablas de forma persistente fuera del contenedor.

- Acceder al contenedor de Hive y conectarse con Beeline.

```

> docker restart hive-server
hive-server
> docker exec -it hive-server netstat -an | grep 10000
tcp        0      0 0.0.0.0:10000 0.0.0.0:*        LISTEN

```

- Crear una tabla con SQL e insertar algunos registros.

```

> docker exec -it hive-server /bin/bash
root@9e29d01785d6:/opt# beeline -u jdbc:hive2://localhost:10000
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/opt/hive/lib/log4j-slf4j-impl-2.6.2.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/opt/hadoop-2.7.4/share/hadoop/common/lib/slf4j-log4j12-1.7.10.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
Connecting to jdbc:hive2://localhost:10000
Connected to: Apache Hive (version 2.3.2)
Driver: Hive JDBC (version 2.3.2)
Transaction isolation: TRANSACTION_REPEATABLE_READ
Beeline version 2.3.2 by Apache Hive
0: jdbc:hive2://localhost:10000> CREATE TABLE IF NOT EXISTS personas (id INT, nombre STRING) ROW FORMAT DELIMITED FIELDS TERMINATED BY ',';
No rows affected (1.899 seconds)
0: jdbc:hive2://localhost:10000> SHOW TABLES;
+-----+
| tab_name |
+-----+
| personas |
+-----+
1 row selected (0.333 seconds)
0: jdbc:hive2://localhost:10000> DESCRIBE personas;
+-----+-----+-----+
| col_name | data_type | comment |
+-----+-----+-----+
| id       | int      |         |
| nombre   | string   |         |
+-----+-----+-----+
2 rows selected (0.18 seconds)
0: jdbc:hive2://localhost:10000>

```

- Consultar los datos con `SELECT * FROM tabla`.

```
No rows affected (1.895 seconds)
0: jdbc:hive2://localhost:10000> SELECT count(*) FROM personas;
+-----+
| _c0 |
+-----+
| 11 |
+-----+
1 row selected (0.506 seconds)
0: jdbc:hive2://localhost:10000> SELECT * FROM personas ORDER BY id;
WARNING: Hive-on-MR is deprecated in Hive 2 and may not be available in the future versions. Consider using a different execution engine (i.e.
spark, tez) or using Hive 1.X releases.
+-----+-----+
| personas.id | personas.nombre |
+-----+-----+
| 1 | Andr??s Palop |
| 2 | Dani Alves |
| 3 | Javi Navarro |
| 4 | Julien Escud?? |
| 5 | David Castedo |
| 6 | Jes??s Navas |
| 7 | Jos?? Luis Mart?? |
| 8 | Enzo Maresca |
| 9 | Adriano Correia |
| 10 | Javier Saviola |
| 11 | Luis Fabiano |
+-----+-----+
11 rows selected (1.535 seconds)
0: jdbc:hive2://localhost:10000>
```

[David Cueli] Comentarios

Utilizando Beeline para ejecutar CREATE y SELECT.

Nota: la base de datos de Hive no está apuntando a HDFS, por lo que no se pide ver los datos desde HDFS, solo realizar las consultas.

Parte 4: Integración con Sqoop

- Crear una base de datos llamada ejemplo en MySQL y una tabla personas.

```
# Parte 4: Integración con Sqoop
db-mysql-metastore:
  image: mysql:5.7
  container_name: db-mysql-metastore
  environment:
    - MYSQL_ROOT_PASSWORD=root
    - MYSQL_DATABASE=sqoop_db
    - MYSQL_USER=user_sqoop
    - MYSQL_PASSWORD=pass_sqoop
  ports:
    # Como tengo WAMP instalado, lo abro en el 3307
    - "3307:3306"

hive-server:
  image: bde2020/hive:2.3.2-postgresql-metastore
  container_name: hive-server
  depends_on:
    - namenode
```

`> docker compose up -d db-mysql-metastore`
[+] up 15/15
✓ Image mysql:5.7 Pulled 26.8s
✓ Container db-mysql-metastore Started 1.1s

~/p/c/r/ai-and-big-data/applied_bigdata/src/activities/lesson_02 on develop !1 ?2 ✓ took 28s applied_bigdata Py at 10:12:51

```
> docker exec -it db-mysql-metastore mysql -u user_sqoop -ppass_sqoop sqoop_db
mysql: [Warning] Using a password on the command line interface can be insecure.
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 2
Server version: 5.7.44 MySQL Community Server (GPL)

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE TABLE fichajes (id INT PRIMARY KEY, jugador VARCHAR(100), temporada VARCHAR(20));
Query OK, 0 rows affected (0.05 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_sqoop_db |
+-----+
| fichajes            |
+-----+
1 row in set (0.00 sec)

mysql>
```

[David Cueli] Comentarios

Creación del contenedor para MySQL (tengo instalado WAMP server en local, en Windows por lo que apunto al puerto 3307) y posterior creación de tabla más adición mínimos datos.

- Insertar algunos registros manualmente.

```
mysql> INSERT INTO fichajes VALUES (1, 'Renato', '2004/05'), (2, 'Kanout', '2005/06'), (3, 'Luis Fabiano', '2005/06');
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM fichajes;
+----+-----+-----+
| id | jugador   | temporada |
+----+-----+-----+
| 1  | Renato    | 2004/05   |
| 2  | Kanout    | 2005/06   |
| 3  | Luis Fabiano | 2005/06   |
+----+-----+-----+
3 rows in set (0.00 sec)

mysql>
```

- Descargar el driver de MySQL desde la web de Oracle:
<https://dev.mysql.com/downloads/connector/j/> (se usó la versión 9.2.0).
- Copiar el archivo mysql-connector-j-9.2.0.jar al contenedor de Sqoop con:
`docker cp mysql-connector-j-9.2.0.jar sqoop:/usr/local/sqoop/lib/`
- Ejecutar el comando Sqoop para importar los datos desde MySQL a HDFS:
`sqoop import --connect jdbc:mysql://mysql:3306/ejemplo \`
`--username root --password root \`
`--table personas \`
`--target-dir /user/root/personas_sqoop \`
`--m 1`

```

$ docker exec -it sqoop sqoop import \
--connect jdbc:mysql://db-mysql-metastore:3306/sqoop_db \
--username user_sqoop \
--password pass_sqoop \
--table fichajes \
--target-dir /user/root/fichajes_sqoop \
--m 1
Warning: /usr/local/sqoop/./hbase does not exist! Hbase imports will fail.
Please set $HBASE_HOME to the root of your Hbase installation.
Warning: /usr/local/sqoop/./hcatalog does not exist! HCatalog jobs will fail.
Please set $HCAT_HOME to the root of your HCatalog installation.
Warning: /usr/local/sqoop/./accumulo does not exist! Accumulo imports will fail.
Please set $ACCUMULO_HOME to the root of your Accumulo installation.
Warning: /usr/local/sqoop/./zookeeper does not exist! Zookeeper imports will fail.
Please set $ZOOKEEPER_HOME to the root of your Zookeeper installation.
2026-05-10 10:17:13,803 INFO sqoop.Sqoop: Running Sqoop version: 1.4.7
2026-05-10 10:17:13,838 WARN tool.BaseSqoopTool: Setting your password on the command-line is insecure. Consider using -P instead.
2026-05-10 10:17:13,957 INFO manager.MySQLManager: Preparing to use a MySQL streaming resultset.
2026-05-10 10:17:13,958 INFO tool.CodeGenTool: Beginning code generation
Loading class 'com.mysql.jdbc.Driver'. This is deprecated. The new driver class is 'com.mysql.cj.jdbc.Driver'. The driver is automatically registered via the SPI and manual loading of the driver class is generally unnecessary.
2026-05-10 10:17:14,720 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM 'fichajes' AS t LIMIT 1
2026-05-10 10:17:14,784 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM 'fichajes' AS t LIMIT 1
2026-05-10 10:17:14,792 INFO orm.CompilationManager: HADOOP_MAPRED_HOME is /usr/local/hadoop
Note: /tmp/sqoop-root/compile/04973112dc34b112d14f2ade40254b91/fichajes.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.
2026-05-10 10:17:16,825 INFO orm.CompilationManager: Writing jar file: /tmp/sqoop-root/compile/04973112dc34b112d14f2ade40254b91/fichajes.jar
2026-05-10 10:17:16,838 WARN manager.MySQLManager: It looks like you are importing from mysql.
2026-05-10 10:17:16,839 WARN manager.MySQLManager: This transfer can be faster! Use the --direct
2026-05-10 10:17:16,839 WARN manager.MySQLManager: option to exercise a MySQL-specific fast path.
2026-05-10 10:17:16,839 INFO manager.MySQLManager: Setting zero DATETIME behavior to convertToNull (mysql)
2026-05-10 10:17:16,846 INFO mapreduce.ImportJobBase: Beginning import of fichajes
2026-05-10 10:17:16,947 INFO Configuration.deprecation: mapred.job.tracker is deprecated. Instead, use mapreduce.job.tracker address
2026-05-10 10:17:17,004 INFO Configuration.deprecation: mapred.jar is deprecated. Instead, use mapreduce.job.jar
2026-05-10 10:17:17,752 INFO Configuration.deprecation: mapred.map.tasks is deprecated. Instead, use mapreduce.job.maps
2026-05-10 10:17:17,826 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-05-10 10:17:18,483 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/root/staging/job_1778407947542_0001
2026-05-10 10:17:27,272 INFO db.DBInputFormat: Using read committed transaction isolation
2026-05-10 10:17:28,150 INFO mapreduce.JobSubmitter: number of splits:1
2026-05-10 10:17:28,192 INFO Configuration.deprecation: yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-05-10 10:17:28,289 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1778407947542_0001
2026-05-10 10:17:28,291 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-05-10 10:17:28,591 INFO conf.Configuration: resource-types.xml not found
2026-05-10 10:17:28,592 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-05-10 10:17:28,893 INFO impl.YarnClientImpl: Submitted application application_1778407947542_0001
2026-05-10 10:17:28,946 INFO mapreduce.Job: The url to track the job: http://a083ff8f139:8088/proxy/application_1778407947542_0001/
2026-05-10 10:17:28,947 INFO mapreduce.Job: Running job: job_1778407947542_0001
2026-05-10 10:17:39,079 INFO mapreduce.Job: Job job_1778407947542_0001 running in uber mode : false
2026-05-10 10:17:39,081 INFO mapreduce.Job: map 0% reduce 0%
2026-05-10 10:17:45,155 INFO mapreduce.Job: map 100% reduce 0%
2026-05-10 10:17:45,166 INFO mapreduce.Job: Job job_1778407947542_0001 completed successfully
2026-05-10 10:17:45,261 INFO mapreduce.Job: Counters: 32
File System Counters
  FILE: Number of bytes read=0
  FILE: Number of bytes written=221544
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=87
  HDFS: Number of bytes written=57
  HDFS: Number of read operations=6
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=2
Job Counters
  Launched map tasks=1
  Other local map tasks=1
  Total time spent by all maps in occupied slots (ms)=3877
  Total time spent by all reduces in occupied slots (ms)=0
  Total time spent by all map tasks (ms)=3877
  Total vcore-milliseconds taken by all map tasks=3877
  Total megabyte-milliseconds taken by all map tasks=3970048
Map-Reduce Framework
  Map input records=3
  Map output records=3
  Input split bytes=87
  Spilled Records=0
  Failed Shuffles=0
  Merged Map outputs=0
  GC time elapsed (ms)=217
  CPU time spent (ms)=4280
  Physical memory (bytes) snapshot=360255488
  Virtual memory (bytes) snapshot=2903347200
  Total committed heap usage (bytes)=810640248
  Peak Map Physical memory (bytes)=360255488
  Peak Map Virtual memory (bytes)=2903347200
File Input Format Counters
  Bytes Read=0
File Output Format Counters
  Bytes Written=57
2026-05-10 10:17:45,267 INFO mapreduce.ImportJobBase: Transferred 57 bytes in 27.5078 seconds (2.0721 bytes/sec)
2026-05-10 10:17:45,271 INFO mapreduce.ImportJobBase: Retrieved 3 records.

```

Parte 5: Script opcional de automatización

Crear un script init.sh que automatice parte del flujo:

- Crear directorios en HDFS
- Subir archivos de texto
- Ejecutar el MapReduce

El script debe ejecutarse con permisos desde el NameNode.

```
#!/bin/bash

echo "====="
echo "   SISTEMA DE AUTOMATIZACIÓN HADOOP   "
echo "====="

# 1. Limpiar rastro de ejecuciones anteriores
echo "[1/7] Limpiando entorno..."
hdfs dfs -rm -r /user/root/automation 2>/dev/null

# 2. Crear estructura de carpetas para el proyecto
echo "[2/7] Creando directorios en HDFS..."
hdfs dfs -mkdir -p /user/root/automation/input

# 3. Crear un archivo de texto grande y subirlo
echo "[3/7] Generando archivo de prueba..."
for i in {1..1000}; do echo "Lorem fistrum pecador a gramenawer fistro pecador"; done >
chiquito_ipsum.txt

echo "[4/7] Subiendo archivo a HDFS..."
hdfs dfs -put -f chiquito_ipsum.txt /user/root/automation/input/

# 4. Verificar que todo está en su sitio
echo "[4/7] Verificando estructura final..."
hdfs dfs -ls -R /user/root/automation

# 5. Ejecutar el trabajo MapReduce (WordCount)
echo "[6/7] Ejecutando MapReduce WordCount..."
hadoop jar /opt/hadoop-3.2.1/share/hadoop/mapreduce/hadoop-mapreduce-examples-3.2.1.jar wordcount
/user/root/automation/input /user/root/automation/logs

echo "====="
echo "[7/7] Proceso finalizado, resultado..."
hdfs dfs -ls -R /user/root/automation/logs
echo "-----"
hdfs dfs -cat /user/root/automation/logs/part-r-00000
echo "-----"
echo "AUTOMATIZACIÓN COMPLETADA"
echo "====="
```

[David Cueli] Comentarios

Para la creación del script ***init.sh*** programo un flujo para:

1. limpiar el entorno (entendiendo se va a ser ejecutado múltiples veces),
2. prepara directorios,
3. sube datos y
4. lanza el MapReduce automáticamente

Optimiza la administración del clúster, garantizando que los procesos de limpieza y ejecución se realicen siempre bajo los mismos parámetros, minimizando errores.

Captura la salida del script confirmando que el trabajo ha finalizado y mostrando los resultados del conteo, certificando así, que la automatización es funcional.

Entrega

Un informe en PDF con:

- Capturas de cada paso realizado
- Explicaciones breves de qué se está haciendo y para qué
- Evidencias de funcionamiento de los servicios y resultados de comandos

Opcional: entregar también el script de automatización